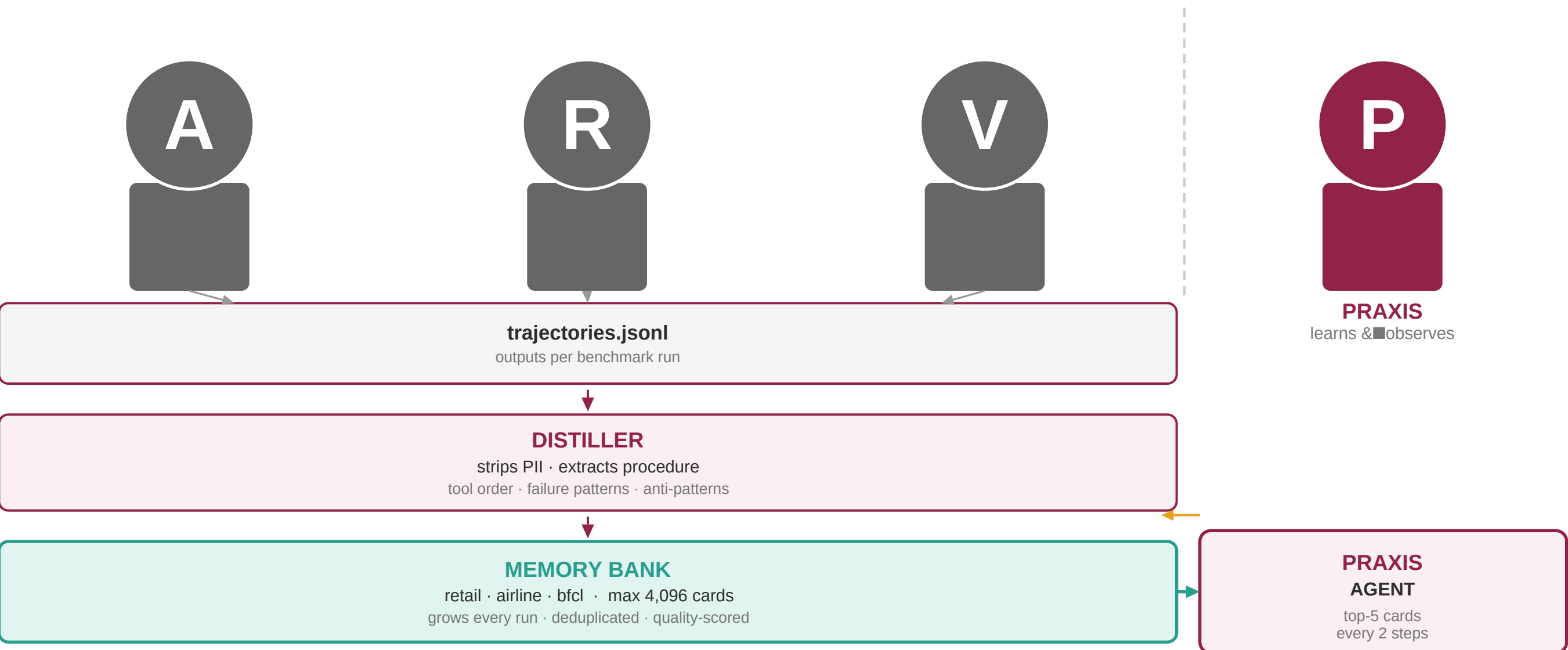


The Core Idea — The Analogy

"An agent that learns from every interaction — not just its own."



After each run, new cards are distilled back — every run starts smarter.

Key Numbers

0

model fine-tuning steps

4

agents compared

5

cards retrieved per query

2

steps between memory refreshes

2,400

max chars in TacticalPlaybook

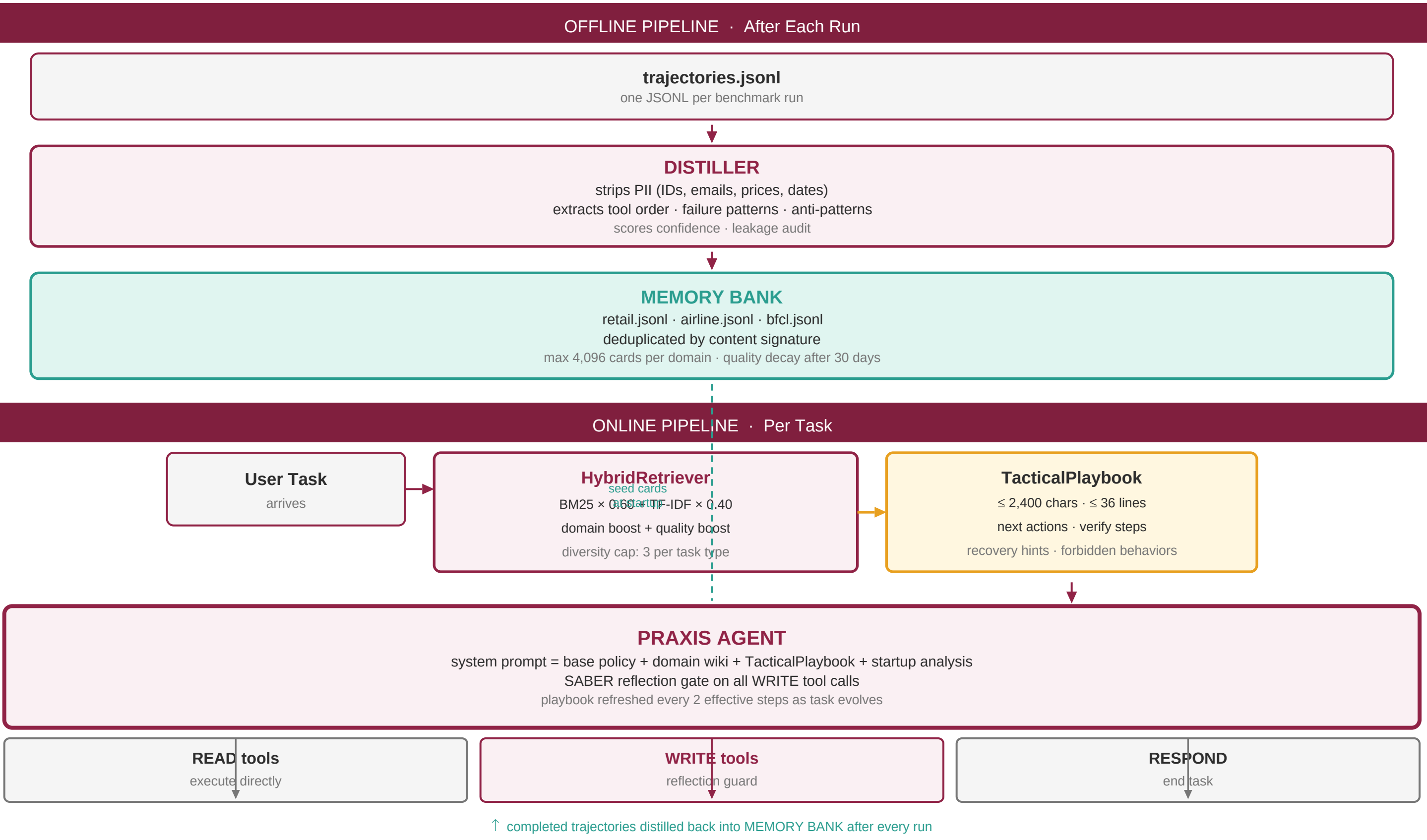
4,096

max memory cards per domain

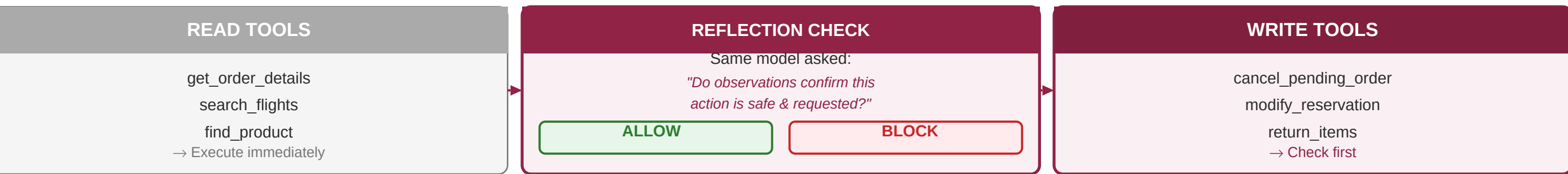
Future Work

1	Cross-Domain Transfer Can retail lessons about 'verify before mutating' help an airline agent? Memory is currently siloed per domain.
2	Active Memory Querying Let PRAXIS explicitly request past failures: 'retrieve cancel_reservation failures' — agent-initiated recall at any point.
3	Mid-Task Online Distillation Distill lessons within a trajectory. Hit failure at step 5, learn and adjust by step 8 without waiting for run to complete.
4	Multi-Agent Shared Memory Multiple PRAXIS instances share a federated memory bank while keeping private task data completely local.

System Architecture



Mutation Reflection Guard (SABER-style)



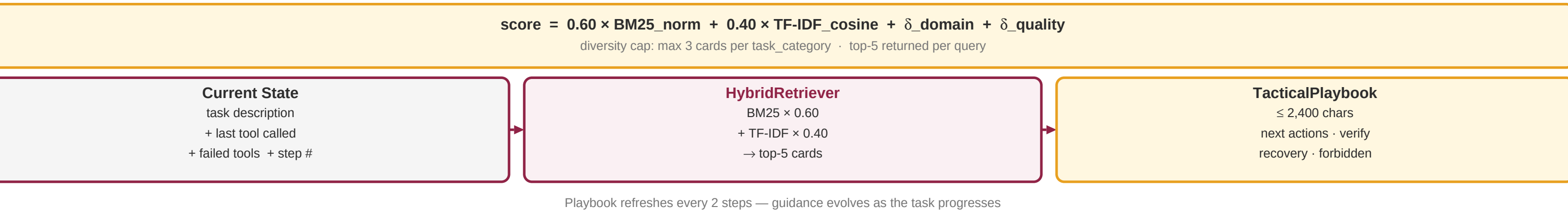
Novel Contributions

Continual Procedural Distillation First system to continuously distill agent trajectories into structured, leakage-audited procedural memory without model retraining.
Leakage-Safe Memory Architecture Raw trajectories are never stored. Multi-layer PII audit. Test-split blocking prevents answer contamination.
Experience from ALL Agents Memory is distilled from ALL controllers (Baseline, Act, ReAct, PRAXIS). Failures become negative-example lessons.

ProcessMemoryCard — What PRAXIS Remembers

ProcessMemoryCard	
[domain: retail · outcome: ✓]	
Task type	exchange_delivered_order_items
Outcome	successful (reward = 1.0)
Lesson	After item lookup: get_order_details → get_product_details → exchange_delivered_order_items
Tool order	[ground_identity, get_order_details, get_product_details, exchange_delivered_order_items]
Don't do	copy IDs, emails, or payment methods from prior examples
Verify	fetch live order state from get_tool before any mutating call
Confirm	ask user to confirm exact target, option, and irreversible effect
Confidence	0.87 · Used: 12+ · Successes: 10 · Failures: 2
× NO order IDs · × NO emails · × NO payment data · × NO prices · × NO raw transcripts Only the procedure. Only the lesson.	

Hybrid Retrieval — Finding the Right Lesson



Experimental Results

Benchmark	Vanilla TC	Act	ReAct	PRAXIS ★	Δ
tau-bench retail	—	—	—	—	+ pp
tau-bench airline	—	—	—	—	+ pp
BFCL V4	—	—	—	—	+ pp

Same model · same tools · same tasks · temperature 0.0 · zero fine-tuning

What Makes PRAXIS Different

Standard RAG retrieves documents to answer questions	PRAXIS retrieves procedures to execute multi-step tasks →
In-Context Learning static examples in every prompt — expensive, fixed	PRAXIS growing compact lessons accumulated across all runs →
Fine-tuning expensive, slow, risks catastrophic forgetting	PRAXIS zero training · works on any frozen OpenAI-compatible model →